

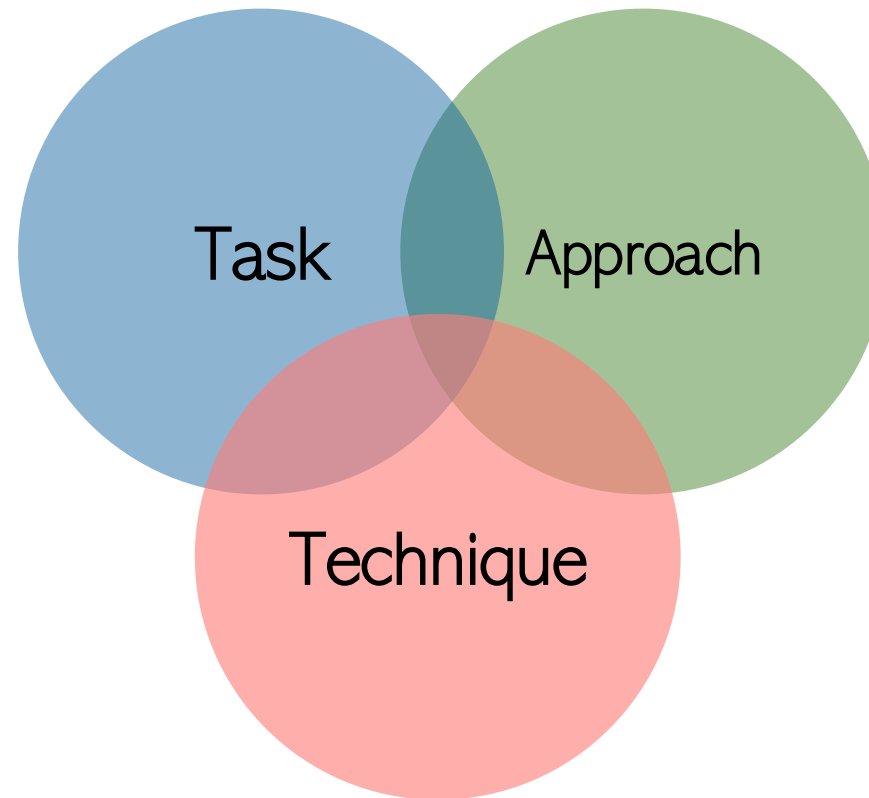


The Oracle of Delphi: How the Ancient Greeks Relied on One Woman's Divine Visions

Outline: W02B: Topics Start

Classification

Regression
Clustering
Ranking (Recommendation)
Vector Representation and
Dimensionality Reduction
Language Modeling



Full/Self/Semi-Supervised Learning

Unsupervised Learning
Reinforcement Learning

k-Nearest Neighbor, Naive Bayes, Logistic Regression
Artificial Neural Networks, Graph Neural Network
Support Vector Machines
Principal Component Analysis
K-Means

4 Outline: W02C: Topics Start

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k -Nearest Neighbor

III) Evaluation

5 Outline: W02C: Topics Start

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k -Nearest Neighbor

III) Evaluation



He Loves Me, He Loves Me Not
By John William Godward
1896, Oil on canvas
Neoclassicism

6 Classification: Definition

Natural Language:

Learning to assign an observation to **one** of a **predefined** set of classes.

Formal Language:

Given a set of world observations $\mathbf{O}$ and a finite set $\mathbf{C}$, the task is to **learn** a mapping function $f: \mathbf{O} \rightarrow \mathbf{C}$, such that $f(o) = c; o \in \mathbf{O}, c \in \mathbf{C}$.

Terminologies:

- Observations $\mathbf{O} \rightarrow$ Experiences $\mathbf{X} \rightarrow$ Data $\mathbf{D}$
- $\mathbf{C}$ is the predefined set of classes (categories)
- c is the class (label) of o
- Boolean Classification: Number of classes, or size of the finite set $\mathbf{C}$, i.e., $|\mathbf{C}| = 2$
- Multi-Class Classification: Number of classes, or size of the finite set $\mathbf{C}$, i.e., $|\mathbf{C}| > 2$

7 Classification: Definition

Natural Language:

Learning to assign an observation to **one** of a **predefined** set of classes.

Formal Language:

Given a set of world observations $\mathbf{X}$ and a finite set $\mathbf{Y}$, the task is to **learn** a mapping function $f: \mathbf{X} \rightarrow \mathbf{Y}$, such that $f(x) = y; x \in \mathbf{X}, y \in \mathbf{Y}$.

Terminologies:

- Observations $\mathbf{O} \rightarrow$ Experiences $\mathbf{X} \rightarrow$ Data $\mathbf{D}$
- $\mathbf{Y}$ is the predefined set of classes (categories)
- y is the class (label) of x
- Boolean Classification: Number of classes, or size of the finite set $\mathbf{Y}$, i.e., $|\mathbf{Y}| = 2$
- Multi-Class Classification: Number of classes, or size of the finite set $\mathbf{Y}$, i.e., $|\mathbf{Y}| > 2$

8

Boolean Classification: Applications

1: being the important class.

Boolean Classes $\mathbf{C} = \mathbf{Y} = \{1, 0\}$

Spam Detection	1: Spam, 0: Not Spam
Fraud Detection	1: Fraudulent, 0: Not Fraudulent (Legitimate)
Patient Diagnosis	1: Disease, 0: No Disease
Sentiment Classification	1: Positive, 0: Not Positive (Negative)
Credit Default Prediction	1: Default, 0: No Default
Phishing Detection	1: Phishing, 0: Not Phishing (Legitimate)
Face Verification	1: Same Person, 0: Not Same Person
Customer Churn Prediction	1: Churn, 0: No Churn (Stay)
Loan Approval	1: Approve, 0: Not Approve (Reject)
Toxicity Detection	1: Toxic, 0: Non-toxic
Intrusion Detection	1: Attack, 0: No Attack (Normal)
Fake-news Detection	1: Fake, 0: Not Fake (Real)
Recommendation Response	1: Click, 0: No Click
...	...

Observation $\mathbf{O} = \mathbf{X}$

Email (text, date, time, sender, ...)
Transaction (account, date, time, ...)
Medical tests (lab results, scans, ...)
Person (face image, text, ...)
Customer information
URL, webpage, email
Two face images
Customer history
Applicant information
Conversation, communication, ...
Network traffic
News article
User, product
...



He Loves Me, He Loves Me Not
By John William Godward
1896, Oil on canvas
Neoclassicism

9

Multi-Class Classification: Applications

Classes $C = Y$

Handwritten Digit Recognition	0, 1, ..., 9
Image Object Classification	Cat, Dog, Horse, Bird, ...
Topic Classification	Politics, Sports, Business, Technology, ...
Language Identification	English, French, Persian, Arabic, ...
Emotion Classification	Joy, Anger, Sadness, Fear, ...
Weather Classification	Sunny, Cloudy, Rainy, Snowy, ...
Medical Diagnosis	Disease A, Disease B, Disease C, Healthy
Document Classification	Invoice, Contract, Resume, Report, ...
Part-of-speech Classification	Noun, Verb, Adjective, Adverb, ...
Music Genre Classification	Rock, Jazz, Classical, Pop, ...
Activity Recognition	Walking, Running, Sitting, Cycling, ...
Network Attack Classification	DoS, Probe, Malware, Credential attack, Normal, ...
...	...

Observation $O = X$

Image
Image
Text
Text
Video
Weather data/image
Medical data
Document
Word
Audio
Sensor data
Network traffic
...



The Good, the Bad and the Ugly
1966, Sergio Leone

10

Multi-Class Classification → Boolean Classification?

What if $f(o)$ can only do Boolean classification ...

The Good, the Bad and the Ugly
1966, Sergio Leone



He Loves Me, He Loves Me Not
By John William Godward
1896, Oil on canvas
Neoclassicism

11

Multi-Class Classification → Multiple Boolean Classification

$\mathbf{C} = \{\text{Good, Bad, Ugly}\}$

$\mathbf{O} = \{\text{Athena, Odysseus, Apollo, Ares, Typhon, ...}\}$

One-vs-Rest

$\mathbf{C}_{\text{Good}} = \{\text{Good, \{Bad, Ugly\}}\} = \{\text{Good, Not Good}\} \rightarrow f(o)$

$\mathbf{C}_{\text{Bad}} = \{\text{Bad, \{Good, Ugly\}}\} = \{\text{Bad, Not Bad}\} \rightarrow g(o)$

$\mathbf{C}_{\text{Ugly}} = \{\text{Ugly, \{Good, Bad\}}\} = \{\text{Ugly, Not Ugly}\} \rightarrow h(o)$

If we have $n = |\mathbf{C}|$ classes, how many different tasks (functions)?



12

Multi-Class Classification → Multiple Boolean Classification

$\mathbf{C} = \{\text{Good, Bad, Ugly}\}$

$\mathbf{O} = \{\text{Athena, Odysseus, Apollo, Ares, Typhon, ...}\}$

One-vs-One

$$\begin{aligned} \mathbf{C}_{\text{Good-Bad}} &= \{\text{Good, Bad}\} \rightarrow f(o) \\ \mathbf{C}_{\text{Good-Ugly}} &= \{\text{Good, Ugly}\} \rightarrow g(o) \\ \mathbf{C}_{\text{Bad-Ugly}} &= \{\text{Bad, Ugly}\} \rightarrow h(o) \end{aligned}$$

If we have $n=|\mathbf{C}|$ classes, how many different tasks (functions)?



Multi-Label Classification: Definition

Natural Language:

Learning to assign an observation to **one** of a **predefined** set of classes.

~~Learning to~~ assign the same observation to **one** of another **predefined** set of classes.

~~Learning to~~ assign the same observation to **one** of another **predefined** set of classes.

...

Formal Language:

Given a set of world observations $\mathbf{O}$ and finite sets $\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \dots$, the task is to **learn** a mapping function $f: \mathbf{O} \rightarrow \mathbf{C}_1 \times \mathbf{C}_2 \times \mathbf{C}_3 \times \dots$, such that $f(o) = (\underbrace{c, c', c'', \dots}_{\text{labels of } o}); o \in \mathbf{O}, c \in \mathbf{C}_1, c' \in \mathbf{C}_2, c'' \in \mathbf{C}_3, \dots$

14

Multi-Label Classification: Example

Lazada Product Title Quality Challenge

https://github.com/hosseinfani/product_title_quality_cikm_2017

O = {Titles in English}
 C_1 : Clarity = {1: Clear, 0: Not Clear}
 C_2 : Conciseness = {1: Concise, 0: Not Concise}



Clarity

"hot sexy red clutch rug sack travel backpack unisex cheap with free gift"

Conciseness

"Hot Sexy Tom Clovers Womens Mens Classy Look Cool Simple Style Casual Canvas Crossbody Messenger Bag Handbag Fashion Bag Tote Handbag Gray"

Why sellers do that?

$f(\text{"Hot Sexy Red Clutch Rug Sack Travel Backpack ..."}) = (0: \text{Not Clear}, 0: \text{Not Concise})$
 $f(\text{"Bold Black Backpack with Striking Red Lips \& Teeth..."}) = (1: \text{Clear}, 0: \text{Concise})$
 $f(\text{"Backpack"}) = (0: \text{Not Clear}, 1: \text{Concise})$
 $f(\text{"Black Backpack with Red Lips Graphic"}) = (1: \text{Clear}, 1: \text{Concise})$

15

Multi-Label Multi-Class Classification: Example

Team Formation: Form Teams of Size 3 for AI Projects

O = {Projects}

C_{1: Grad} = {Reia, Ali, Luiz, ...} → Leader

C_{2: Undergrad} = {Meisam, Sara, Mahdis, Ahmed, ...} → Designer

C_{3: Undergrad} = {Meisam, Sara, Mahdis, Ahmed, ...} → Developer

C_{4: Branch} = {Victoria, Montreal, ...} → Location

$f(\text{"Study the effects of CSC501xSENG747 Course Projects on Global Warming"}) = (\text{Ali, Mahdis, Ahmed, Victoria})$

$f(\text{"Build an Autonomous Car"}) = (\text{Reia, Meisam, Sara, Montreal})$

16

Multi-Label Multi-Class Classification: Example

Why not multiple multi-class classifications?

$f(\text{"Study the effects of CSC501xSENG747 Course Projects on Global Warming"}) = (\text{Ali, Mahdis, Ahmed, Victoria})$

→

$f(\text{"Study the effects of CSC501xSENG747 Course Projects on Global Warming"}) = (\text{Ali})$

$g(\text{"Study the effects of CSC501xSENG747 Course Projects on Global Warming"}) = (\text{Mahdis})$

$h(\text{"Study the effects of CSC501xSENG747 Course Projects on Global Warming"}) = (\text{Ahmed})$

$i(\text{"Study the effects of CSC501xSENG747 Course Projects on Global Warming"}) = (\text{Victoria})$

$f(\text{"Build an Autonomous Car"}) = (\text{Reia, Meisam, Sara, Montral})$

→

$f(\text{"Build an Autonomous Car"}) = (\text{Reia})$

$g(\text{"Build an Autonomous Car"}) = (\text{Meisam})$

$h(\text{"Build an Autonomous Car"}) = (\text{Sara})$

$i(\text{"Build an Autonomous Car"}) = (\text{Montral})$

Yes, it is doable, but not interested, because ...

17

Multi-Label Multi-Class Classification: Example

Why not multiple multi-class classifications?

$f(\text{"Study the effects of CSC501xSENG747 Course Projects on Global Warming"}) = (\text{Ali, Mahdis, Ahmed, Victoria})$

→

$f(\text{"Study the effects of CSC501xSENG747 Course Projects on Global Warming"}) = (\text{Ali})$

$g(\text{"Study the effects of CSC501xSENG747 Course Projects on Global Warming"}) = (\text{Mahdis})$

$h(\text{"Study the effects of CSC501xSENG747 Course Projects on Global Warming"}) = (\text{Ahmed})$

$i(\text{"Study the effects of CSC501xSENG747 Course Projects on Global Warming"}) = (\text{Victoria})$

$f(\text{"Build an Autonomous Car"}) = (\text{Reia, Meisam, Sara, Montral})$

→

$f(\text{"Build an Autonomous Car"}) = (\text{Reia})$

$g(\text{"Build an Autonomous Car"}) = (\text{Meisam})$

$h(\text{"Build an Autonomous Car"}) = (\text{Sara})$

$i(\text{"Build an Autonomous Car"}) = (\text{Montral})$

- Learning many functions,
- But more importantly, ...

18

Multi-Label Multi-Class Classification

→ Multiple Multi-Class Classification? Yes.

→ Multiple Multiple Boolean Classification? Yes.

But not interested, because:

- Learning many functions,
- But more importantly, missing **correlations!**

[Engn-announce] Masters Non-Thesis Oral Examination - Ayushi Pandey - September 21, 2026



Engn-announce <engn-announce-bounces@lists.uvic.ca> on behalf of Victoria Simpson <vsimpson@uvic.ca>

To: ENGR Announce <engn-announce@lists.uvic.ca>



Mon 14/09/2026 12:42

Department of Electrical and Computer Engineering

Master's Non-Thesis Oral Examination

Candidate: Ayushi Pandey

Project: "Malware Categorization Using Deep Learning for Multi-Class Classification."

Date: Monday, September 21

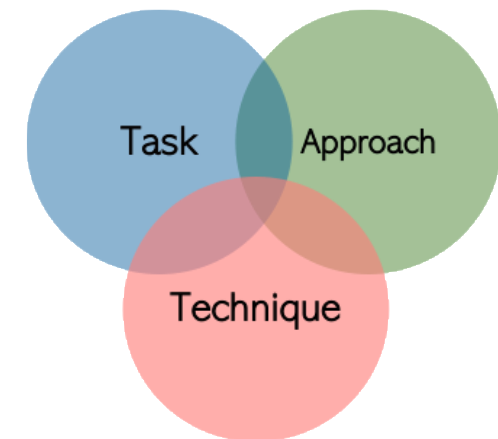
Time: 1:00 pm (PDT)

Place: Online, via Zoom

Please contact the supervisor for information on attending the remote defense.

Supervisor(s): Dr. Issa Traoré, Dr. Isaac Woungang

Warm regards,
Victoria



Outline: W02C: Topics Start

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k -Nearest Neighbor

III) Evaluation

Aletheia or Alethia /æliˈθaɪ.ə/ is truth or disclosure in philosophy. Originating in Ancient Greek philosophy, used for the first time in by Parmenides in his poem On Nature. He contrasts it with doxa (opinion). In Greek mythology, Aletheia was personified as a Greek goddess of truth, sincerity and unconcealment.

"the daughter of Zeus, unforgetting Truth"

